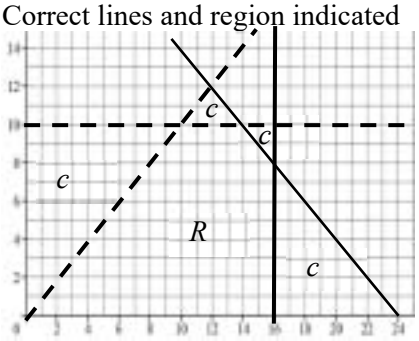
| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 6(b)(i) | $180-(26+42)$ | **B1** | |
| 6(b)(ii) | 6.89 or 6.888 to 6.892... | **3** | **M2** for $\frac{9.55}{\sin 112}\times\sin 42$ oe<br>or **M1** for $\frac{\sin 112}{9.55}=\frac{\sin 42}{CD}$ oe |
| 6(c) | 5.84[2…] | **3** | **M2** for $\frac{x}{6.5}=\sin 64$ oe<br><br>or **M1** for identifying shortest distance from $D$ is perpendicular to $AB$ |
| 7(a) | 2 | **2** | **M1** for $3x+4x=6+8$ or better |
| 7(b) | $5a(2a+1)$ final answer | **2** | **B1** for $a(10a+5)$ or $5(2a^2+a)$<br>or $5a(2a+1)$ then spoilt |
| 7(c) | $4x(x-3)$ final answer | **2** | **M1** for $((2x-3)-3)((2x-3)+3)$ or better<br>or for $4x^2-6x-6x+9$ [–9] oe or better |
| 7(d)(i) | $\frac{1}{15}$ oe | **1** | |
| 7(d)(ii) | 19 683 | **2** | **B1** for g(9), $3^9$ or $3^{3^x}$ seen |
| 7(d)(iii) | –3 | **2** | **M1** for $3^k=\frac{1}{27}$ or $3^k=3^{-3}$<br>or answer g(–3) |
| 8(a) | $y<10$<br>$y<x$ oe<br>$x+y\leqslant 24$ oe | **3** | **B1** for each<br><br>If 0 scored, **SC1** for $y\leqslant 10$ and $y\leqslant x$ and $x+y<24$ |

| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 8(b) | Correct lines and region indicated   | **6** | **B1** for each correct line<br><br>and<br><br>**B2** for R in correct region for all 4 correct lines<br><br>or **B1** for R in any one of the regions marked *c*<br><br>or **B1** for R that satisfies 3 of the correct inequalities |
| 8(c) | 228 nfww | **2** | **M1** for $8x + 12y$ for any $(x, y)$ in *their* R, $x, y$ both integer<br>or $x = 15$, $y = 9$ |
| 9(a) | [$a$ =] 9<br><br>[$b$ =] 14 | **3** | **B2** for $a$ =9<br>OR<br>**M2** for<br>$\frac{60}{360}\times 2\times\pi\times 17+\frac{60}{360}\times 2\times\pi\times 10+7+7$<br>oe<br>or **M1** for<br>$\frac{60}{360}\times 2\times\pi\times 17$ oe **or** $\frac{60}{360}\times 2\times\pi\times 10$ oe<br><br>If 0 scored **SC1** for $b$ =14 |
| 9(b)(i) | 60° at centre<br>or interior angle = 120° | **B1** | |
| | [6×]$\frac{1}{2}\times d^2\times\sin 60$ oe | **M1** | |
| | [$d^2$ =]$\frac{127.3}{6\times\frac{1}{2}\times\sin 60}$ | **M1** | |
| | 6.99[9…] to 7.00[…] | **A1** | Dep on M1M1 |
| 9(b)(ii)(a) | 1273 | **1** | |
| 9(b)(ii)(b) | 675 or 674.5 to 674.6 | **2** | **M1** for 2 ×127.3 oe or 6 × 7 × 10 oe |
| 10(a)(i) | (4.5, −1) | **2** | **B1** for each |